Thinking Through a Space with a Personal AI

Research interests and initial PhD proposal

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Start at the entrance

Imagine walking into a small immersive exhibition with an AI assistant on your glasses. At the entrance, I point to an introductory panel: "If we turn this towards the doorway, what will visitors see first?" The assistant shows two positions. I walk towards the projection wall and reconsider: "I want them to read the title before noticing the projection. Keep this route open." It compares views from where I stand and updates the digital model as we settle on a layout.

On the next visit, a projector has moved. I want to pick up yesterday's discussion: remember why we placed the panel there, check the changed room, and work out what to keep. This is where I would begin studying a personal AI that can think through a place with its user.

Why I want to build this

My starting point is architectural work, from field records and CAD/Rhino models to lighting and projection at the Damo Cave exhibition. Working on site put physical space and digital systems in the same workflow. I want an AI I can talk things through with while looking at the place itself. My longer-term ambition resembles Jarvis: an assistant that comes to know my work, offers its own judgment, and helps make things happen. XR gives that collaboration a place around me.

Why Synteraction

TOM offers a wearable-assistant platform that connects information about the wearer and environment to assistance [1]. CRAFT carries real-world encounters into a creative project and preserves its context across sessions [2]. PersonaMail supports later, context-sensitive use of communication preferences [3]. This is the part of Heads-Up Computing I want to explore: an assistant that keeps up while someone is looking around and doing the work.

Building on speech-driven AR authoring such as ImaginateAR [4], I would study how an assistant follows a designer through a changing room and revisits the reasons behind a layout on a later visit.

Two questions I would investigate

1. As a person moves and talks, how can an AI connect words such as "here" and "this panel" to the intended place, and help compare what different layouts would mean there?

2. When the person returns, how can the assistant use the reasons behind earlier choices to revise the layout for a changed room?

Build around one room

I would first observe designers walking through a small exhibition space, then review recordings with them to understand which observations lead them to change a layout. The prototype would use a measured room model with a movable panel and a defined projection area. A camera-equipped XR headset would provide the user's view and position. Pointing would identify an object; short spatial previews would show alternatives in place. The person could continue the conversation while moving between viewpoints.

The assistant would call a small set of tools to move or rotate the digital panel, compare views and save a layout version. Each choice would retain the relevant object, the observation point and the person's stated reason: for example, "read the title before seeing the projection". On a return visit, it could revisit that reason against the new equipment positions. The person could correct a reference or change their mind in ordinary conversation, and see the resulting model update.

Teach it what a person means

I also want to train or adapt models for this interaction. A first learning task would be resolving references: which object does "this one" mean when the person has changed position? Walkthrough recordings would pair speech and pointing with the object the person confirms, recording corrections and resulting digital-scene edits. I would compare an existing vision-language model with a task-adapted version, testing on held-out viewpoints and participants. Reference accuracy and time to respond would show whether adaptation helps the conversation keep moving.

Test it while people work

The main comparison would use the same model, environmental inputs and editing tools. Both assistants would be able to retrieve the conversation, saved layout versions, viewpoints and stated reasons, and offer previews, corrections and undo. The experimental assistant would additionally link each layout choice to its observation point and stated reason, and let the person revisit those links when the room changes. This tests whether organizing the history around design decisions helps beyond simply having the conversation available.

Participants with spatial design experience would work through equivalent layout briefs, then return after an equipment change. I would measure mistaken object references and the effort needed to correct them. Geometric checks on the room model could assess route obstructions and title visibility from specified viewpoints; independent designers could assess how well each solution meets the brief. The briefs would identify requirements that still apply after the change. I would score whether participants preserve these and revise the affected choices, recording mistaken carry-over. Decision time and interruptions to looking around the room would show the cost of the interaction. Observation and pilot results would guide the study size and session spacing.

What I bring to the work

Alongside spatial work, I have built a Pygame teammate that intercepts projectiles and signals its actions through text and sound. My machine-learning coursework involved training and comparing classifiers; in an eight-person capstone, I implemented a direct-detection baseline and investigated model errors. I want to bring these skills into an XR system that can take part in a real design conversation. Repeated site visits offer a practical first step towards the personal AI I want to build.

References

[1] Nuwan Janaka et al. (2024). TOM: A Development Platform For Wearable Intelligent Assistants. UbiComp Companion 2024. https://doi.org/10.1145/3675094.3678382

[2] Runze Cai et al. (2026). CRAFT: Exploring Wearable Creative AI on Smart Glasses for Fiction Writing in Real-World Contexts. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 10(3), Article 81. https://doi.org/10.1145/3831952

[3] Rui Yao et al. (2026). PersonaMail: Learning and Adapting Personal Communication Preferences for Context-Aware Email Writing. IUI 2026. https://doi.org/10.1145/3742413.3789123

[4] Jaewook Lee et al. (2025). ImaginateAR: AI-Assisted In-Situ Authoring in Augmented Reality. UIST 2025. https://doi.org/10.1145/3746059.3747635